

Port Adjustment to Maritime Chokepoint Disruption: Evidence from the Red Sea Shipping Crisis

Abstract

The 2023-2024 Red Sea crisis provides a quasi-natural experiment for studying port adjustment under a major maritime chokepoint disruption. We assemble a port-month panel for 175 ports across three shipping corridors, combining IMF PortWatch activity data, UNCTAD's Port Liner Shipping Connectivity Index, IHO basin definitions, and a stylized geographic rerouting-penalty measure based on great-circle distance via Suez versus via the Cape of Good Hope. We estimate continuous-intensity event-study difference-in-differences and triple-difference models with port, month, constituent-IHO-basin-by-month fixed effects, and port-specific linear trends. Coefficients describe within-basin exposure-gradient slopes in shifted, winsorized, port-specific same-month standardized outcomes. Adjustment is corridor-specific. Higher exposure is associated with lower relative imports in the Southeast Asia gateway corridor. Borderline evidence also points to higher relative port calls in the Red Sea-Gulf-western Indian Ocean corridor, while the pooled export gradient is positive but suggestive. Network connectivity shapes some short-run port-call gradients but does not consistently buffer or amplify geographic exposure. The findings caution against summarizing chokepoint disruptions with a single port-level effect.

Keywords: Maritime chokepoint disruption; Geographic rerouting-penalty gradient; Red Sea crisis; Continuous-intensity difference-in-differences; Shipping network connectivity; Event study

1. Introduction

At the end of 2023, escalating security incidents in the Red Sea disrupted a strategic maritime corridor. Major container carriers suspended Suez Canal routings and diverted services around the Cape of Good Hope. The rerouting lengthened Asia-Europe voyages by about 30%, or roughly 10 to 14 days, and tightened effective fleet capacity (Dunn & Leibovici, 2024; UNCTAD, 2024a). It also raised freight costs and schedule uncertainty, adding pressure to global supply chains (Dunn & Leibovici, 2024; Kamali et al., 2024; UNCTAD, 2024a). This paper asks how the disruption translated into port-level changes in calls, imports, and exports, and whether those adjustments differed systematically across ports.

When a maritime chokepoint is disrupted, the shock does not hit ports uniformly. In this paper, we separate three empirically distinct margins: a stylized geographic rerouting-penalty proxy, a liner-network connectivity measure, and cargo/service-specific outcome categories. The first margin defines the continuous exposure gradient; the second is used as a heterogeneity dimension in triple-difference (DDD) models; and the third decomposes outcomes across PortWatch cargo/service categories. These measures do not directly observe realized Red Sea cargo exposure or pre-crisis cargo shares.

Maritime transport carries more than 80% of global merchandise trade by volume, but its efficiency depends on a concentrated network of chokepoints (UNCTAD, 2024a). Disruptions at corridors such as the Suez Canal, the Panama Canal, and the Strait of Malacca can raise trade costs, delay deliveries, and transmit stress through shipping networks and supply chains (Attinasi et al., 2024; UNCTAD, 2024a).

Ports connect maritime and inland supply chains, so corridor disruption appears in port

performance through changes in calls, imports, and exports. The resilience literature treats post-shock performance trajectories as central evidence, including peak loss, recovery speed, and cumulative loss over the disruption window (Holling, 1996; Bruneau et al., 2003; Balbi et al., 2025; Gu et al., 2024). This paper does not construct a resilience index. It estimates standardized port outcome gradients around the Red Sea shock.

Existing studies have confirmed the systematic nature and cascading effects of route disruptions, including diversion costs, shipping-schedule uncertainty, and network reconfiguration (Notteboom et al., 2024). Reports and early assessments of the Red Sea disruption show a significant decline in traffic through the Bab-el-Mandeb Strait and the Suez Canal, while traffic on the route around the Cape of Good Hope has increased, with potentially far-reaching effects on trade and supply chains (UNCTAD, 2024a; World Bank, 2025).

Port-level empirical evidence remains limited in two respects. First, high-frequency quasi-experimental evidence is still scarce on how calls, imports, and exports changed across differently exposed ports. Second, many exposure measures use a single dimension, such as diversion distance, route share, or connectivity. This makes it difficult to separate physical rerouting costs from network position in port adjustment.

We study these issues using the Red Sea crisis as a quasi-natural experiment with a clear onset and cross-port variation in a pre-shock geographic rerouting-penalty proxy. We combine daily port performance indicators from the International Monetary Fund (IMF) PortWatch system with port-level liner connectivity information from the United Nations Conference on Trade and Development (UNCTAD) Port Liner Shipping Connectivity Index (PLSCI) (Hoffmann et al., 2024; International Monetary Fund, 2025; UNCTAD, 2025). Geographic exposure is a stylized rerouting penalty based on substituting a Cape route for a Suez route under

a Europe-oriented counterfactual. Network exposure is measured by the UNCTAD PLSCI averaged over 2019Q1-2023Q4 and is used cautiously as a baseline network-position measure because 2023Q4 overlaps the event quarter. We then estimate static and dynamic exposure gradients in port calls, imports, and exports, and test whether those gradients vary with baseline liner connectivity.

First, we estimate continuous exposure-gradient differences in port adjustment. Rather than comparing treated and untreated gateways, we assign each of 175 exact-match corridor ports a stylized geographic rerouting-penalty measure and estimate how port-call, import, and export outcomes vary along that gradient within constituent-IHO-basin-by-month cells.

Second, we test whether baseline liner-network position conditions the geographic gradient. A continuous triple-difference interacts the rerouting-penalty proxy with PLSCI-based network exposure. This specification tests heterogeneity in the exposure gradient. The results show network conditioning only at specific margins, especially short-run port calls in one corridor, rather than a consistent buffer or amplifier.

Third, we decompose the aggregate gradients across 18 cargo and service outcome subcategories. These exploratory regressions show where aggregate gradients are concentrated across outcome categories.

2. Literature Review

2.1. Shipping route disruptions and deviation

From a mechanism perspective, route disruption is a network-wide shock transmitted through vessel deviation, effective-capacity tightening, and trade adjustment. Higher route risk or

reduced corridor capacity can increase sailing distance and transit time, raise fuel and chartering costs, and tighten effective shipping capacity through longer voyage cycles (UNCTAD, 2024a; Verschuur et al., 2025). These changes can reduce schedule reliability and trigger blank sailings, port skipping, and service reconfiguration. At the port level, the consequences may appear as fewer calls, lower import or export activity, longer waiting times, or congestion, but also as relative gains for substitute gateways that absorb diverted traffic (Yap & Yang, 2024; Kim et al., 2025).

These responses are therefore unlikely to be uniform. Even under the same corridor disruption, some ports may experience contraction while others receive reallocated services or cargo, depending on their corridor position, cargo profile, and access to alternative network connections (Liu et al., 2026; Yap & Yang, 2024).

These responses are also likely to vary systematically with geographic exposure to deviation. When a corridor disruption forces vessels onto longer alternative routes, ports facing a larger diversion burden may be more exposed to higher transport costs, longer voyage cycles, and tighter service capacity (UNCTAD, 2024a; Yap & Yang, 2024; Verschuur et al., 2025).

At the port level, this may translate into sharper adjustments in port calls, trade activity, congestion, or service configuration, including a greater likelihood of service thinning or port skipping (Yap & Yang, 2024). Corridor disruption therefore generates not only an overall network shock, but also a continuous geographic exposure gradient across ports.

Beyond this geographic exposure gradient, the role of network connectivity in shaping port adjustment remains contested. One view emphasizes redundancy and rerouting capacity: ports embedded in more connected networks may have access to a broader set of alternative services and paths, allowing carriers to reallocate capacity and schedules more quickly when a corridor

becomes unavailable. In this view, higher connectivity can strengthen resilience through substitution and path redundancy (Wan et al., 2023; Viljoen & Joubert, 2016).

A contrasting view highlights hub dependence and congestion propagation. Highly connected ports often occupy core positions in liner shipping networks, where flows and coordination complexity are concentrated. When shocks force large-scale rerouting, these hubs may face sharper congestion or transmit disruption more widely through cascading failure dynamics. From this perspective, connectivity can amplify rather than mitigate the effects of corridor disruption (Xu et al., 2022; Zhang et al., 2026).

Taken together, geographic exposure provides a first-order source of heterogeneity, while network connectivity may either buffer or amplify that exposure. The role of connectivity is therefore best understood as an empirical question, which motivates designs that interact route exposure with network-based measures of port position to trace heterogeneous and dynamic performance responses.

2.2. Empirical identification of route-disruption effects

Event-study and difference-in-differences (DID) designs are among the main empirical approaches used to identify the effects of shipping disruptions on port and trade outcomes. In a binary DID setting, researchers define treatment and control groups and compare outcome variables before and after the shock. Causal interpretation relies on the parallel-trends assumption and researchers often control for unit and time fixed effects. More recent work has extended this logic to continuous treatment intensity, which is especially relevant for corridor disruptions because exposure often varies by degree rather than by simple treatment status (Li et al., 2023; Heiland et al., 2025). In port-related applications, exposure intensity may be defined by

pre-shock connectivity, route dependence, additional deviation distance, or insecurity along relevant sea lanes. Existing continuous-intensity DID studies mainly follow two strategies. One uses disruption-induced distance increases as the “dose,” exploiting deviation-induced changes in sea distance or sailing time within a fixed-effects framework (Feyrer, 2021). The other treats insecurity as the “dose,” interacting time-varying piracy or attack intensity with predetermined route exposure to identify differential changes over time in trade or shipping outcomes (Burlando et al., 2015; Besley et al., 2015).

From a causal-inference perspective, binary DID is useful for identifying average post-shock differences, while continuous-intensity DID is better suited for tracing how outcomes vary along exposure gradients. Yet weak or insignificant average gradients do not automatically imply either full resilience or misspecification of the empirical design; they may also mask offsetting responses across ports. A DDD design that adds a second pre-determined exposure dimension, such as network connectivity, provides a way to test whether the geographic exposure gradient is systematically conditioned by port position in the shipping network. In this sense, DDD is best understood as a heterogeneity design that refines the interpretation of DID estimates rather than as a standalone test of resilience. Such applications remain limited in causal studies of corridor disruptions and the Red Sea crisis.

Econometric work on the Red Sea crisis is still limited. Existing studies nevertheless show that corridor risk can be translated into measurable route exposure and that high-frequency maritime data can be used to identify shock-driven changes in trade and shipping outcomes. At the same time, existing literature mainly focuses on trade flows, chokepoint traffic, or average route-level outcomes, and often operationalizes exposure along a single dimension, such as route dependence, distance, or insecurity. What remains less developed is a port-level analysis of how

a corridor shock translates into dynamic adjustments in calls, imports, and exports across differently exposed ports. In particular, existing studies provide limited evidence on whether these adjustments follow a continuous geographic exposure gradient and on whether pre-crisis network connectivity systematically conditions that gradient. These limitations motivate a port-level empirical design that traces dynamic adjustment across ports with different deviation exposure and network positions.

This paper connects three strands of research. Studies of route disruptions document fewer calls, lower throughput, and relative gains for substitute gateways (Yap & Yang, 2024; Kim et al., 2025). We extend this evidence by estimating a continuous exposure gradient in a unified port-month panel. Performance-based resilience studies trace post-shock trajectories (Gu et al., 2024; Balbi et al., 2025). We examine standardized outcome trajectories and exposure-gradient differences. Work on network position often treats connectivity as a static resilience indicator (Liu et al., 2022; Park & Medda, 2015). We instead interact geographic exposure with baseline connectivity to test whether the gradient varies with network position.

3. Data and Methodology

The design treats the Red Sea crisis as an exogenous corridor shock to the liner shipping system. The identifying comparison relies on cross-port variation in a pre-shock geographic rerouting-penalty proxy within a fixed continuous-intensity difference-in-differences specification. A continuous-intensity triple-differences specification tests whether a baseline network-connectivity measure conditions the geographic exposure gradient.

3.1. Data and variables

The analysis uses a port-month panel from four sources: IMF PortWatch daily port activity,

a basin lookup derived from IHO Sea Areas, UNCTAD quarterly PLSCI, and PortWatch port coordinates. PortWatch provides daily port calls and estimated import and export tonnage. The basin lookup assigns ports to corridor groups. PLSCI provides the baseline network measure, and coordinates are used to compute the rerouting-penalty proxy. The monthly panel begins in 2019 for pre-period standardization, but the estimating sample is limited to a fixed event window around 2023m12. After basin filtering and exact matching to PLSCI labels, the event-window sample contains 2,975 port-month observations for 175 ports. After excluding $k = 0$ during panel construction and $k = -1$ at the regression stage, the effective estimating sample contains 2,800 observations before outcome-specific missingness.

3.1.1. Outcome variables

The main analysis focuses on three aggregate monthly outcomes, namely port calls, import tonnage, and export tonnage, and applies the same transformation to the 18 cargo and service subcategories used in the heterogeneity analysis. Daily PortWatch series are first aggregated to the port-month level. Each outcome o is then standardized relative to the port-specific distribution for the same calendar month over the pre-crisis period from 2019m1 to 2023m11. In other words, each port-month observation is benchmarked against that port's own pre-crisis seasonal norm for the same calendar month rather than against the port's overall annual average:

$$z_{it}^o = \frac{Y_{it}^o - \mu_{i,m(t)}^{o,pre}}{\sigma_{i,m(t)}^{o,pre}}, \quad \text{pre} = 2019\text{m}1\text{-}2023\text{m}11 \quad (1)$$

This same-month standardization removes regular within-port seasonality and makes the transformed outcomes comparable across ports of different scales. A constant shift is then applied so that the transformed series remains strictly positive. In the implemented code, the shift for each outcome is the amount needed to raise the minimum z-score to 0.001 when that

minimum is below 0.001. The shifted series is then winsorized at the 1st and 99th percentiles. These transformations are used only for estimation stability and do not change the interpretation that larger values indicate activity above the port's own pre-crisis seasonal norm.

3.1.2. Geographic exposure

Geographic exposure captures the deviation penalty implied by a shift from the Suez Canal to the Cape of Good Hope. Using the Suez Canal, the Cape of Good Hope, and the Strait of Gibraltar as reference points, two stylized routes to the European trunk corridor are constructed for each port (Fig. 1). The two route lengths are defined as follows:

$$viaSuez_i = d(i, Suez) + d(Suez, Ref), \quad viaCape_i = d(i, Cape) + d(Cape, Ref) \quad (2)$$

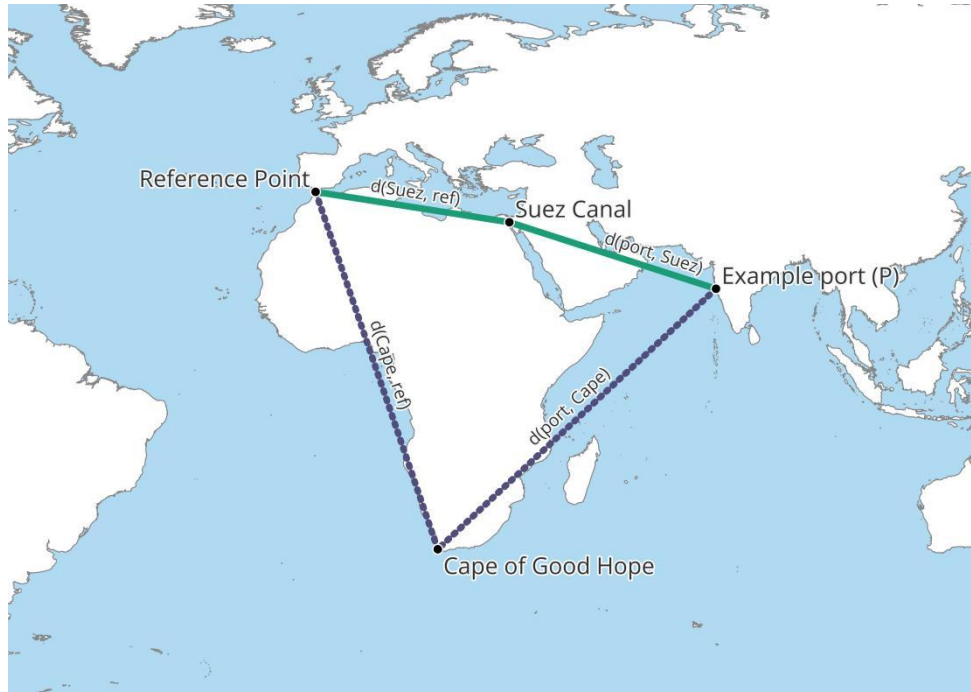


Fig. 1. Construction of the geographic rerouting-penalty measure. For an example port P , two stylized routes to a European reference point near the Strait of Gibraltar are compared: a via-Suez route of length $d(port, Suez) + d(Suez, ref)$ and a via-Cape route of length $d(port, Cape) + d(Cape, ref)$. The raw rerouting penalty is the non-negative excess of the via-Cape route length over the via-Suez route length, following Eqs. (2)–(3). Distances are great-circle approximations and the common reference leg cancels under within-sample standardization.

The raw exposure measure is then defined as:

$$E_{geo,raw,i} = \max\{viaCape_i - viaSuez_i, 0\} \quad (3)$$

Negative raw values are truncated at zero. The positive part of the distribution is then z-score standardized and shifted, while ports with zero raw exposure retain the value 0. The final regression variable is therefore a non-negative transformed measure of the deviation penalty, with larger values indicating stronger dependence on the Suez corridor and a larger rerouting cost under diversion around the Cape.

Algebraically, $viaCape_i - viaSuez_i = [d(i,Cape) - d(i,Suez)] + [d(Cape,Ref) - d(Suez,Ref)]$, where the bracketed reference-leg difference is a common constant (about 4,611 km in the implemented coordinates). Among ports with positive clipped raw exposure, this constant cancels under z-standardization. However, the final variable is not globally equivalent to $d(i,Cape) - d(i,Suez)$, because the code first truncates negative raw values to zero, standardizes only the positive part, shifts the positive part, and keeps zero-exposure ports at 0. The resulting variable is a non-negative geographic rerouting-penalty gradient under a Europe-oriented counterfactual.

3.1.3. Network exposure

Network exposure is measured with the UNCTAD PLSCI and fixed at the baseline average over 2019Q1–2023Q4:

$$E_{net,i} = |Q_{pre}|^{-1} \sum_{q \in Q_{pre}} PLSCI_{iq} \quad (4)$$

PLSCI is merged to the port panel using a conservative exact-match procedure. Because 2023Q4 overlaps the event quarter, the current baseline should be interpreted as a network-

position measure rather than a strictly pre-treatment covariate unless a pre-event-only version excluding 2023Q4 is reported. Before estimation, the port-level average is standardized across matched ports and shifted to remain strictly positive. Higher values indicate stronger integration into the liner shipping network.

3.1.4. Sample definition

The baseline sample retains only ports in these three groups and only those ports that can be exactly matched to PLSCI records (Fig. 2).

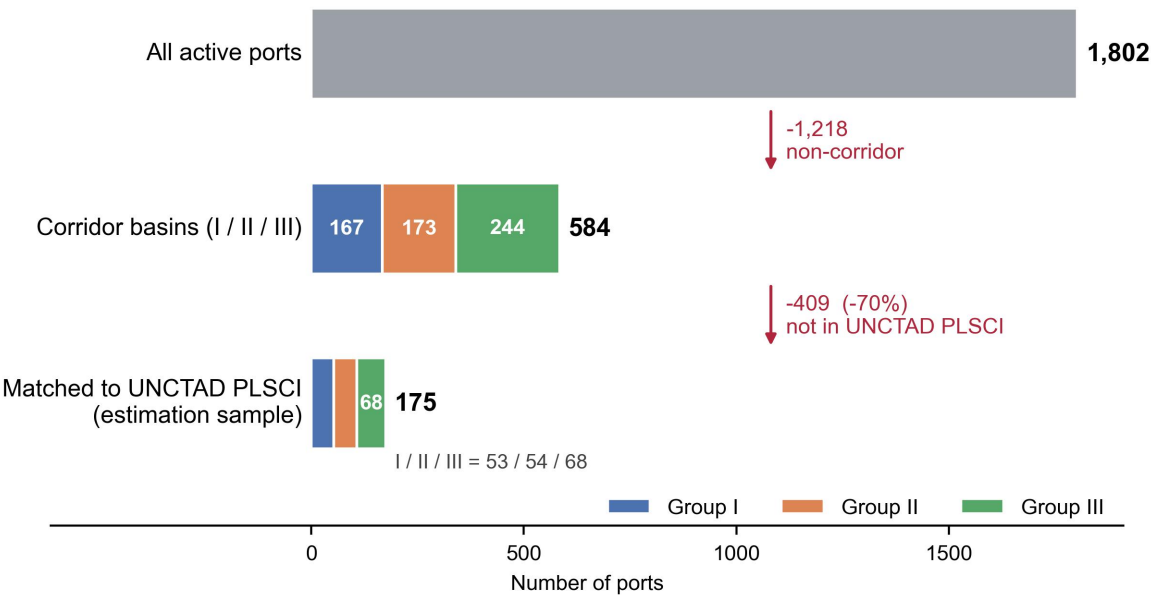


Fig. 2. Sample construction and selection. Of 1,802 active ports, 584 lie in the three corridor basins (I/II/III); 175 exactly match to UNCTAD PLSCI and form the estimation sample. The 409 corridor ports dropped at the UNCTAD merge are not matched under the implemented exact rule. Matched ports are roughly 3.7x the median size of dropped ports, so the estimation sample is restricted to larger, better-connected corridor ports. Details are reported in Table A.2.

3.2. Empirical strategy

Treatment intensity varies continuously across ports, so the crisis is modeled as an exposure gradient rather than a binary treated-control contrast. The event month is December 2023, so $k_{i,t} = t - 2023m12$. The fixed event window is $k \in [-5, 12]$. The code removes $k = 0$ during panel construction and removes $k = -1$ at the regression stage. Post-period coefficients therefore use $k = 1, \dots, 12$, with $k = -2$ as the omitted reference period in dynamic specifications. All models include port fixed effects, month fixed effects, constituent-IHO-basin-by-month fixed effects, and port-specific linear trends.

3.2.1. Continuous-intensity event-study DID

The baseline static DID estimates the post-period geographic exposure gradient, while the event-study version traces its timing. The static specification is:

$$Y_{it} = \beta_1(E_{geo,i} \times Post_t) + \beta_2(E_{net,i} \times Post_t) + \alpha_i + \lambda_t + \theta_{b(i),t} + \gamma_i Trend_t + \varepsilon_{it} \quad (5)$$

The corresponding dynamic DID replaces the single post indicator with event-time interactions after excluding $k = 0$ and $k = -1$:

$$Y_{it} = \alpha_i + \lambda_t + \theta_{b(i),t} + \gamma_i Trend_t + \sum_{k \in K} [\beta_k D_{kt} E_{geo,i} + \eta_k D_{kt} E_{net,i}] + \varepsilon_{it}, \quad K = \{-5, -4, -3, 1, \dots, 12\} \quad (6)$$

The coefficient β_k measures the month- k slope with respect to geographic exposure relative to $k = -2$. Differential pre-trends are assessed by a joint test over $k = -5, -4, -3$, and post-period differences are summarized by a joint test over $k = 1, \dots, 6$. The month-by-month confidence intervals reported below are pointwise intervals rather than simultaneous bands.

3.2.2. Continuous-intensity event-study DDD

The DDD specifications add baseline network exposure as a second heterogeneity dimension.

264 The static DDD specification is:

$$Y_{it} = \beta_1(E_{geo,i} \times Post_t) + \beta_2(E_{geo,i} \times E_{net,i} \times Post_t) + \beta_3(E_{net,i} \times Post_t) + \alpha_i + \lambda_t + \theta_{b(i),t} + \gamma_i Trend_t + \varepsilon_{it} \quad (7)$$

265 The dynamic DDD specification is:

$$Y_{it} = \alpha_i + \lambda_t + \theta_{b(i),t} + \gamma_i Trend_t + \sum_{k \in K} [\beta_k D_{kt} E_{geo,i} + \gamma_k D_{kt} E_{net,i} + \delta_k D_{kt} E_{geo,i} E_{net,i}] + \varepsilon_{it} \quad (8)$$

266 The parameter of interest is δ_k . The marginal geographic effect at event time k is:

$$\frac{\partial Y_{it}}{\partial E_{geo,i}} = \beta_k + \delta_k E_{net,i} \quad (9)$$

267 When lower outcomes indicate larger losses, $\delta_k > 0$ is consistent with attenuation of the
 268 adverse geographic gradient for more connected ports, whereas $\delta_k < 0$ is consistent with
 269 amplification. Thus, δ_k captures conditional variation in the geographic gradient.

270 3.2.3. Estimation and inference

271 All models are estimated as linear regressions after residualizing port, month, and
 272 constituent-IHO-basin-by-month fixed effects and including port-specific linear trends. The same
 273 baseline and dynamic specifications are estimated for the pooled exact-match sample and
 274 separately for corridor Groups I, II, and III. The aggregate tables focus on the three main
 275 outcomes, while parallel regressions are also run for 18 cargo/service subcategories.

276 Standard errors are clustered at the port level. The same covariance structure is used for joint
 277 pre-trend tests and post-period joint tests. The main robustness checks re-estimate the baseline
 278 results after dropping the single largest port in the estimation pre-window and after replacing the
 279 baseline exposure scaling with raw standardized E_{geo} and E_{net} measures. Placebo evidence is
 280 based on 200 treatment-strength randomizations of geographic exposure across ports. A reported

randomization value of 0.000 should be read as 0/200 exceedances under the unadjusted empirical rule, not as an exact zero p-value. A time-placebo design is retained as appendix evidence because the pre-event window is relatively short.

Outcome coefficients are expressed in shifted, winsorized units of a port's own pre-crisis same-calendar-month standard deviation. One baseline geographic-exposure unit corresponds to one standard deviation of the positive clipped rerouting-penalty distribution, about 2,000 km in the exact-match sample. Because zero-exposure ports are retained at 0 and positive values are shifted after standardization, the final geographic-exposure regressor is not mean-zero with unit standard deviation over all ports.

4. Results and Analysis

4.1. Baseline effects and corridor heterogeneity

Throughout, the coefficient on $E_{geo} \times post$ is a within-basin exposure-gradient effect: with port, month, and constituent-IHO-basin-by-month fixed effects, it measures how much more (or less) geographically exposed ports adjusted relative to less-exposed ports in the same constituent basin and month, rather than an average effect of the crisis on port activity. The pooled ('ALL') estimate is a single common-slope pooled regression over the exact-match three-corridor sample rather than a precision-weighted average of corridor-specific slopes. The triple-difference term $E_{geo} \times E_{net} \times post$ asks whether this geographic gradient varies with baseline connectivity.

Table 1 reports the aggregate baseline estimates discussed in the text. Three findings stand out: a positive pooled within-basin exposure-gradient (DID) estimate for exports, a positive Basin I DID effect for port calls, and a negative Basin II DID effect for imports.

At the pooled level, the clearest raw-p result appears on the export margin. The DID coefficient on $E_geo \times post$ is 0.730 ($p = 0.022$), indicating that, after the onset of the Red Sea disruption, ports with higher geographic exposure recorded relatively stronger standardized export activity than less exposed ports within comparable basin-month cells. Because the dependent variable is standardized relative to each port's own pre-crisis seasonal norm and then shifted and winsorized, the estimate should be read as a relative change in standardized export activity rather than as a uniform increase in raw export volumes. However, neither pooled port calls nor pooled imports yields a comparably robust DID effect, and the pooled DDD coefficient for exports is insignificant. The pooled result is therefore concentrated on the export margin.

Table 1. Baseline aggregate estimates discussed in the text.

Sample	Outcome	Effect	Estimate	Pre-trend p-value	Drop-top1 p-value	Raw-standardized p-value	Exposure placebo p-value
Pooled (ALL)	Export	Geographic-exposure effect (DID)	0.730** (0.315)	0.989	0.021	0.019	0.000
Basin I	Port calls	Geographic-exposure effect (DID)	1.034** (0.422)	0.887	0.018	0.020	0.000
Basin I	Export	Geographic-exposure effect (DID)	0.726* (0.403)	0.943	0.078	0.075	0.000
Basin II	Import	Geographic-exposure effect (DID)	-1.385*** (0.449)	0.529	0.002	0.003	0.000
Basin II	Port calls	Network-moderation effect (DDD)	0.324*** (0.082)	0.137	0.009	0.000	0.070

*Notes: Entries report the aggregate main-outcome estimates discussed in the text. DID denotes the coefficient on $E_geo \times Post$ and DDD denotes the coefficient on $E_geo \times E_net \times Post$. Robust standard errors are shown in parentheses. Pre-trend p-value reports the test for differential pre-shock trends. Drop-top1 p-value reports the p-value from a re-estimation that excludes the largest pre-shock port in the estimation pre-window. Raw-standardized p-value reports the p-value from a re-estimation using alternatively standardized exposure measures. Exposure placebo p-value reports the randomization-based p-value from reassigning exposure intensity across ports; 0.000 means 0/200 randomized assignments exceeded the observed estimate under the unadjusted empirical rule. Stars denote raw cluster-robust p-values: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$.*

The basin split shows that the pooled estimate conceals different adjustment margins. In Basin I, the strongest result is a positive DID effect for port calls (1.034, $p = 0.018$). This coefficient suggests that the crisis was associated with a reallocation of vessel activity toward more geographically exposed ports in the Red Sea, Gulf, and western Indian Ocean corridor. Basin I also shows a positive export DID estimate, but it is only marginally significant (0.726, $p = 0.078$) and is therefore treated as supplementary rather than core evidence on the export margin. Overall, the Basin I pattern is primarily operational, with export strengthening as a secondary margin of adjustment.

Basin II presents a different pattern. The DID coefficient for imports is negative and strongly significant (-1.385 , $p = 0.003$), implying that higher geographic exposure is associated with a sharper post-shock decline in import activity. At the same time, the positive Basin II port-calls DDD estimate suggests that network position conditioned how this exposure translated into operational adjustment within the basin. The estimate remains significant in the baseline specification and under both re-estimation checks, but the weaker exposure-placebo diagnostic means that it is best treated as supplementary rather than core evidence on network-conditioned operational adjustment.

The basin decomposition clarifies the pooled result. The positive pooled export DID coefficient coexists with a Basin I pattern centered on operational reallocation and a Basin II pattern centered on import compression. This contrast is central to interpreting the pooled estimate.

Given the number of estimated gradients, the aggregate results are interpreted using a formal multiple-testing hierarchy. We apply the Benjamini-Hochberg procedure (Benjamini & Hochberg, 1995) to the 24 confirmatory main-outcome tests (four samples, three aggregate

outcomes, and two effects). The Basin II import DID remains significant at the 5% FDR level ($q = 0.039$). The Basin II port-calls DDD also remains below the 5% FDR threshold ($q = 0.003$), but its weaker exposure-placebo diagnostic keeps it in the supplementary category. The pooled export DID and Basin I port-calls DID are borderline at the 10% FDR level (both $q \approx 0.087$), and the Basin I export DID does not survive the correction ($q \approx 0.218$). Accordingly, Basin II import is the core aggregate finding; Basin II port-calls DDD is FDR-robust but supplementary; and the pooled export and Basin I port-calls DID estimates are interpreted as borderline or suggestive.

4.2. Dynamic adjustment patterns

The event-study results add a temporal dimension to the baseline estimates by showing when the post-shock gradients emerge. Table 2 summarizes the pointwise timing patterns for the discussed aggregate estimates, while Fig. 3 plots the Basin I port-calls DID, Basin II import DID, and pooled export DID paths together with the Basin II port-calls DDD path, which is retained as a supplementary interaction effect.

Table 2. Event-time profile of the discussed results.

Sample	Outcome	Effect	Pre-event significant months	Post-event significant months	First post-event month
Pooled (ALL)	Export	DID	None	3, 6	3
Basin I	Port calls	DID	None	3, 5	3
Basin I	Export	DID	None	6	6
Basin II	Import	DID	None	1, 5, 7	1
Basin II	Port calls	DDD	None	1	1

Notes: The dynamic summary is based on the event-study specification. The pre-event and post-event columns list event months in which the coefficient is significant at the 5% level using pointwise confidence intervals.

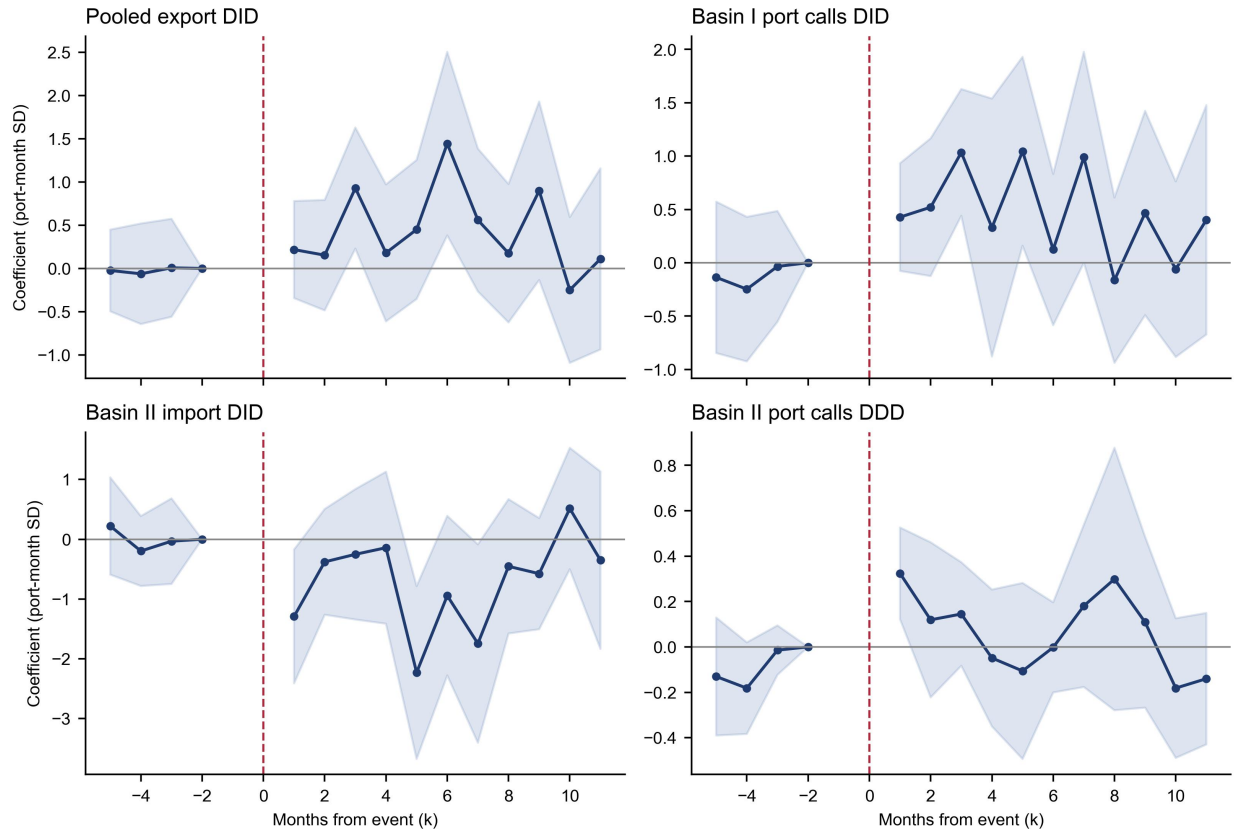


Fig. 3. Selected dynamic adjustment paths. Panel A: pooled export DID; Panel B: Basin I port-calls DID; Panel C: Basin II import DID; Panel D: Basin II port-calls DDD. Shaded bands indicate pointwise 95% confidence intervals; the dashed vertical line marks the event month (December 2023, $k = 0$).

The pooled export DID path is episodic in the pointwise event-study estimates. The coefficient is statistically significant at $k = 3$ and again at $k = 6$, indicating selected-horizon rather than continuous export adjustment as carriers changed routes, schedules, and capacity after the disruption.

Basin I port calls show pointwise post-period signals at $k = 3$ and $k = 5$, suggesting a delayed and non-continuous port-call response rather than a contemporaneous one-month spike. By contrast, the Basin I export DID effect is thinner in the dynamic specification, becoming significant only at $k = 6$. The dynamic evidence therefore supports treating Basin I port calls as

the primary operational margin and Basin I exports as supplementary.

Basin II imports exhibit the strongest pointwise contraction pattern among the discussed paths. The DID coefficient is significant at $k = 1$ and reappears at $k = 5$ and $k = 7$, suggesting an early but non-continuous import contraction. In contrast, the Basin II port-calls DDD signal is concentrated in the short run: its dynamic significance is confined to $k = 1$, consistent with a transient network-conditioned reshuffling of vessel activity rather than a persistent reordering of the basin's port hierarchy.

Overall, the dynamic evidence should be read as pointwise timing evidence rather than as a simultaneous path test. The pooled export gradient is selective and episodic; Basin I port calls show delayed, non-continuous post-period signals; Basin II imports show an early and recurrent but non-continuous contraction pattern; and the Basin II port-calls DDD appears short-lived. These temporal differences indicate that the reported results reflect different margins of adjustment across corridors rather than a single common dynamic mechanism.

4.3. Cargo-type heterogeneity and network-conditioned responses

Fig. 4 summarizes the aggregate coefficient landscape, plotting the three main outcomes across the four corridor groups for both the DID and DDD estimators in a single heatmap. It consolidates the headline estimates discussed in Sections 4.1–4.2 — the negative Basin II import DID, the positive Basin II port-calls DDD, and the more marginal pooled export and Basin I port-calls DID effects — and the complete decomposition into cargo and service types is reported for every outcome in Appendix Fig. A.1 (DID) and Fig. A.2 (DDD).

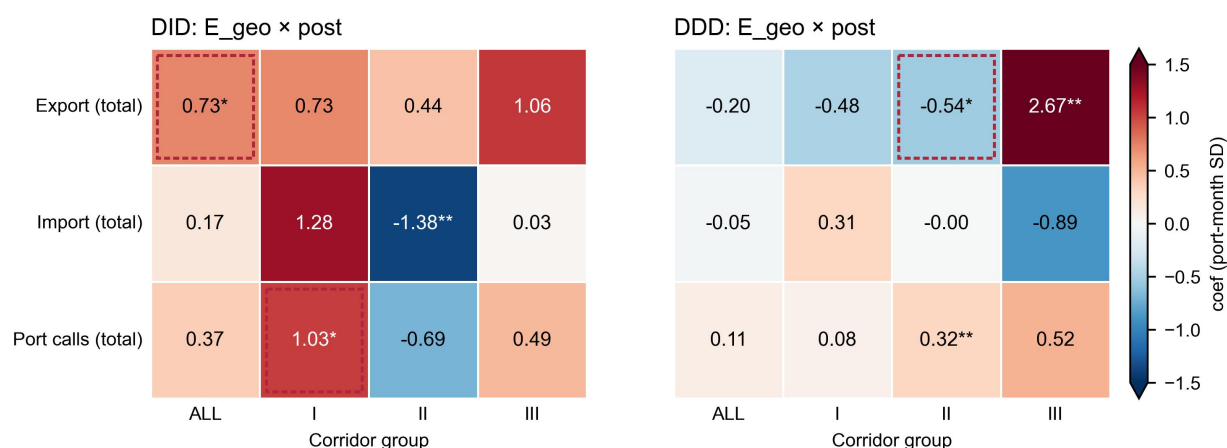


Fig. 4. Coefficient heatmap of the aggregate estimates. Each cell reports the estimated coefficient (in units of a port's own pre-crisis monthly standard deviation) for one corridor group (columns: ALL, I, II, III) and one main outcome (rows: exports, imports, port calls). The left panel shows the geographic exposure-gradient difference-in-differences ($E_{geo} \times post$; DID) and the right panel the connectivity-conditioned triple difference ($E_{geo} \times E_{net} \times post$; DDD). Asterisks denote Benjamini–Hochberg FDR-adjusted significance as described in Section 4.1 (** $q < 0.05$, * $q < 0.10$); a dashed red border marks cells that are significant at the raw 5% level but do not survive the FDR correction. Red denotes positive and blue negative coefficients; the color scale is capped at ± 1.5 standard deviations.

The 18 cargo/service subtype regressions are exploratory decompositions that show where aggregate gradients are concentrated; they do not identify pre-crisis cargo mix as a causal moderator. Table 3 reports selected subtype estimates that help interpret the aggregate and dynamic results.

Table 3. Exploratory cargo-type and service-type outcome decomposition.

Sample	Subcategory	Outcome	Effect	Estimate	Pre-trend p-value	Coefficient p-value
<i>Panel A. Pooled evidence</i>						
Pooled (ALL)	dry bulk	Export	Geographic-exposure effect (DID)	1.022** (0.491)	0.556	0.039
Pooled (ALL)	general cargo	Port calls	Network-moderation effect (DDD)	0.239** (0.104)	0.470	0.023
Pooled (ALL)	roro	Import	Network-moderation effect	-0.467*** (0.126)	0.163	<0.001

Sample	Subcategory	Outcome	Effect	Estimate	Pre-trend p-value	Coefficient p-value
			(DDD)			
Pooled (ALL)	cargo	Port calls	Network-moderation effect (DDD)	0.208* (0.122)	0.125	0.090
Pooled (ALL)	container	Port calls	Network-moderation effect (DDD)	0.195* (0.111)	0.363	0.081
Panel B. Basin I evidence						
Basin I	dry bulk	Port calls	Network-moderation effect (DDD)	0.573** (0.234)	0.210	0.018
Basin I	roro	Export	Geographic-exposure effect (DID)	5.621*** (1.445)	0.421	<0.001
Basin I	cargo	Port calls	Geographic-exposure effect (DID)	0.838* (0.426)	0.265	0.055
Basin I	tanker	Import	Geographic-exposure effect (DID)	1.406* (0.751)	0.834	0.067
Panel C. Basin II evidence						
Basin II	cargo	Import	Geographic-exposure effect (DID)	-1.625*** (0.402)	0.833	<0.001
Basin II	dry bulk	Import	Geographic-exposure effect (DID)	-0.884* (0.509)	0.873	0.089
Basin II	cargo	Port calls	Network-moderation effect (DDD)	0.381*** (0.117)	0.488	0.002
Basin II	container	Port calls	Network-moderation effect (DDD)	0.491** (0.186)	0.641	0.011
Basin II	cargo	Export	Network-moderation effect (DDD)	-0.475** (0.236)	0.468	0.049
Basin II	dry bulk	Export	Network-moderation effect (DDD)	-0.806** (0.338)	0.128	0.021
Panel D. Basin III evidence						
Basin III	roro	Export	Geographic-exposure effect (DID)	-2.385*** (0.637)	0.091	<0.001
Basin III	container	Port calls	Network-moderation effect (DDD)	1.849*** (0.696)	0.335	0.010

Notes: Entries report selected estimates from the 18 cargo/service subtype outcomes, with robust standard errors in parentheses. These estimates are used as exploratory outcome decompositions for interpreting the aggregate results reported in Sections 4.1 and 4.2. Stars denote raw cluster-robust p-values and are not FDR-adjusted: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. The Basin III export entry in Panel D refers to the RORO export subtype DID, not to the aggregate export DDD.

At the pooled level, the positive aggregate export DID estimate is concentrated in dry-bulk exports, where the DID coefficient is 1.022 ($p = 0.039$). This indicates that the pooled export gradient is relatively narrow in outcome-category terms. The pooled subtype results also reveal selected network-conditioned adjustments on the service side: general-cargo port calls show a

positive DDD effect (0.239, $p = 0.023$), while RORO imports show a negative DDD effect (-0.467, $p < 0.001$). Taken together, the pooled subtype picture is one of a concentrated export response accompanied by narrower service-side adjustments.

Basin I's subtype pattern is more specific than the aggregate coefficients alone suggest. The positive basin-level port-calls DID result is complemented by a positive dry-bulk port-calls DDD estimate (0.573, $p = 0.018$), suggesting that part of the operational response varied with network position and geography. The clearest trade-side signal within Basin I is the large positive DID coefficient for RORO exports (5.621, $p < 0.001$). The subtype evidence is therefore consistent with Basin I adjustment being primarily operational, with trade-side strengthening concentrated in selected cargo categories.

Basin II presents the clearest subtype pattern among the selected decompositions. The aggregate import contraction is mirrored by a negative cargo-import DID coefficient (-1.625, $p < 0.001$) and a weaker dry-bulk import DID coefficient (-0.884, $p = 0.089$). This suggests that the basin-level import gradient is concentrated in selected import categories. On the service side, the positive aggregate port-calls DDD result is reflected in cargo port calls (0.381, $p = 0.002$) and container port calls (0.491, $p = 0.011$).

4.4. Robustness and diagnostic checks

Table 4 reports the baseline p-values together with three diagnostics for the aggregate findings discussed in the text: the drop-top1 re-estimation, the raw-standardized exposure re-estimation, and the exposure-placebo test.

Table 4. Robustness diagnostics for the aggregate findings discussed in the text.

Sample	Outcome	Effect	Baseline p-value	Drop-top1 p-value	Raw-standardized p-value	Exposure placebo p-value
Pooled (ALL)	Export	DID	0.022	0.021	0.019	0.000
Basin I	Port calls	DID	0.018	0.018	0.020	0.000
Basin I	Export	DID	0.078	0.078	0.075	0.000
Basin II	Import	DID	0.003	0.002	0.003	0.000
Basin II	Port calls	DDD	0.000	0.009	0.000	0.070
Basin II	Export	DDD	0.018	0.059	0.018	0.040

Notes: Entries summarize the retained robustness checks and exposure-placebo diagnostics for the aggregate results most relevant to the discussion. Lower placebo p-values indicate that randomized exposure assignments are less able to reproduce the observed estimate. Values of 0.000 indicate 0/200 exceedances under the unadjusted empirical rule, not an exact zero p-value.

The three DID findings discussed above remain stable across these checks. The pooled export exposure-gradient (DID) estimate, the Basin I port-calls DID estimate, and the Basin II import DID estimate all remain statistically significant when the largest pre-shock port is excluded and when the baseline exposure variables are replaced by the raw-standardized alternative. In the 200 exposure randomizations, none of the randomized assignments equaled or exceeded the observed estimates in absolute value for these three DID findings. These diagnostics suggest that the results are not driven solely by a single dominant port, by one particular scaling choice for exposure, or by exposure assignments that are easily reproduced under random reshuffling.

The supplementary findings are more mixed. Basin I export DID remains significant at the 10% level under the retained checks and is therefore kept as supplementary evidence. Basin II port-calls DDD also remains significant in the baseline specification and under both re-estimation checks. The only diagnostic that does not meet the 5% threshold is the exposure-placebo test ($p = 0.070$), so it is retained as supplementary rather than core evidence on network-conditioned operational adjustment. Overall, the robustness analysis supports a cautious

empirical hierarchy rather than a uniform set of strong findings.

5. Discussion

The central implication is that the Red Sea crisis did not produce a single system-wide port response in the exact-match corridor sample. The basin-level results show that different margins dominate in different corridors. The pooled export DID coefficient captures a within-basin differential in relative export activity between more- and less-exposed ports. Basin I is characterized mainly by operational reallocation in port calls, while Basin II is characterized by contraction in import activity. This interpretation is consistent with recent work on the Red Sea disruption, which documents uneven deviation effects, port-call reconfiguration, and corridor-specific consequences rather than a uniform decline across ports and activities (Notteboom et al., 2024; World Bank, 2024; Yap & Yang, 2024).

The evidence on network moderation is more nuanced than a broad DDD narrative would suggest. The supplementary Basin II port-calls DDD result suggests that baseline network position is associated with how exposure translated into short-run operational adjustment within that basin. However, the absence of equally strong and robust DDD effects across the rest of the aggregate outcome matrix indicates that the role of connectivity varied by adjustment margin. This interpretation fits the wider shipping-network literature, which shows that alternative deviation and connectivity can improve substitution in some cases while also intensifying cascading congestion and dependence in others (Wan et al., 2023; Xu et al., 2022).

The cargo/service results reinforce this interpretation. The pooled export gradient is concentrated in dry-bulk exports; Basin I combines operational adjustment with selective strengthening in RORO exports; and Basin II combines import weakness with supplementary

evidence of network-sensitive reallocation in cargo and container port calls. The results differ across cargo/service outcome categories rather than showing a common shift across all activities. For transport and logistics analysis, route geometry alone is not sufficient to summarize observed adjustment; outcome categories and service roles must also be considered.

The policy implication is that vulnerability assessment should not rely on the pooled-sample gradient alone. A planner who focused only on the positive pooled export gradient would miss the fact that one corridor showed mainly service reallocation while another showed a clear decline in import activity. Likewise, a strategy based only on physical rerouting costs would overlook the limited but relevant role of network position in some operational adjustments. For port authorities and carriers, the results suggest mapping vulnerability separately across trade-volume exposure, service-reallocation exposure, and network-role exposure.

The research design sets clear boundaries for interpretation. The estimates capture relative differences within the exact-match corridor sample, not responses of all ports worldwide. The exact-match design improves measurement alignment between PortWatch and PLSCI for matched ports, but it narrows external validity because the retained ports are larger and better connected than the typical corridor port. The current network-exposure baseline averages PLSCI through 2023Q4, which overlaps the event quarter; network-moderation findings therefore require caution unless a pre-event-only sensitivity excluding 2023Q4 is added. The design could be adapted to other chokepoints only with episode-specific rerouting-penalty measures and comparable port-level exposure data. The paper documents heterogeneous exposure-gradient adjustment in this Red Sea episode; it does not demonstrate universal port resilience or a portable Cape-versus-Suez mechanism.

Data and code availability

The replication code, derived analysis files, and table/figure source files needed to reproduce the reported results will be deposited in a public repository upon publication. The analysis uses IMF PortWatch daily port activity data and UNCTAD PLSCI data from the original providers. Raw source data are redistributed only where permitted by the providers' terms of use.

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work, the authors used Claude (Anthropic) and AI-assisted editorial audit tools to improve language, readability, terminology consistency, and typesetting. The authors reviewed and edited all AI-assisted changes and take full responsibility for the content of the publication.

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594 **Appendix A. Basin-group composition**

595 **Table A.1.** Basin-group composition in the final analysis panel.

Constituent basin	Ports	Observations
<i>Panel A. Basin I (11 constituent basins; 53 ports; 901 observations)</i>		
Andaman or Burma Sea	1	17
Arabian Sea	8	136
Bay of Bengal	6	102
Gulf of Aden	4	68
Gulf of Aqaba	1	17
Gulf of Oman	1	17
Gulf of Suez	1	17
Indian Ocean	12	204
Laccadive Sea	3	51
Persian Gulf	11	187
Red Sea	5	85
<i>Panel B. Basin II (16 constituent basins; 54 ports; 918 observations)</i>		
Arafura Sea	2	34
Bali Sea	2	34
Banda Sea	3	51
Celebes Sea	1	17
Flores Sea	1	17
Gulf of Thailand	3	51
Gulf of Tomini	1	17
Java Sea	2	34
Makassar Strait	3	51
Malacca Strait	4	68
Molukka Sea	1	17
Savu Sea	1	17
Singapore Strait	2	34
South China Sea	22	374

Constituent basin	Ports	Observations
Sulu Sea	5	85
Timor Sea	1	17
Panel C. Basin III (6 constituent basins; 68 ports; 1156 observations)		
Eastern China Sea	16	272
Japan Sea	16	272
Philippine Sea	11	187
Sea of Okhotsk	1	17
Seto Naikai or Inland Sea	14	238
Yellow Sea	10	170

Notes: Entries summarize the composition of the final analysis panel used in the exact-match baseline. Observations are port-month observations in the final analysis panel. Port counts denote unique matched ports retained in the final analysis panel.

Table A.2. Sample-selection funnel by corridor group: ports kept versus dropped at the UNCTAD PLSCI merge.

Corridor group	Corridor ports	Matched (kept)	Dropped	Retention	Median size kept	Median size dropped	Size ratio
I (Red Sea–Gulf–W Indian Ocean)	167	53	114	31.7%	894	216	4.1×
II (SE Asia gateway)	173	54	119	31.2%	860	427	2.0×
III (East Asia origin)	244	68	176	27.9%	1543	323	4.8×
All corridor	584	175	409	30.0%	1153	314	3.7×

Notes: Size denotes total vessel count (PortWatch). Dropped ports are corridor ports that were not matched under the implemented exact PLSCI label match. Because PLSCI covers a smaller set of mostly larger, container-capable ports, the estimation sample should be interpreted as a coverage-driven exact-match sample of larger and better-connected corridor ports rather than as all corridor ports.

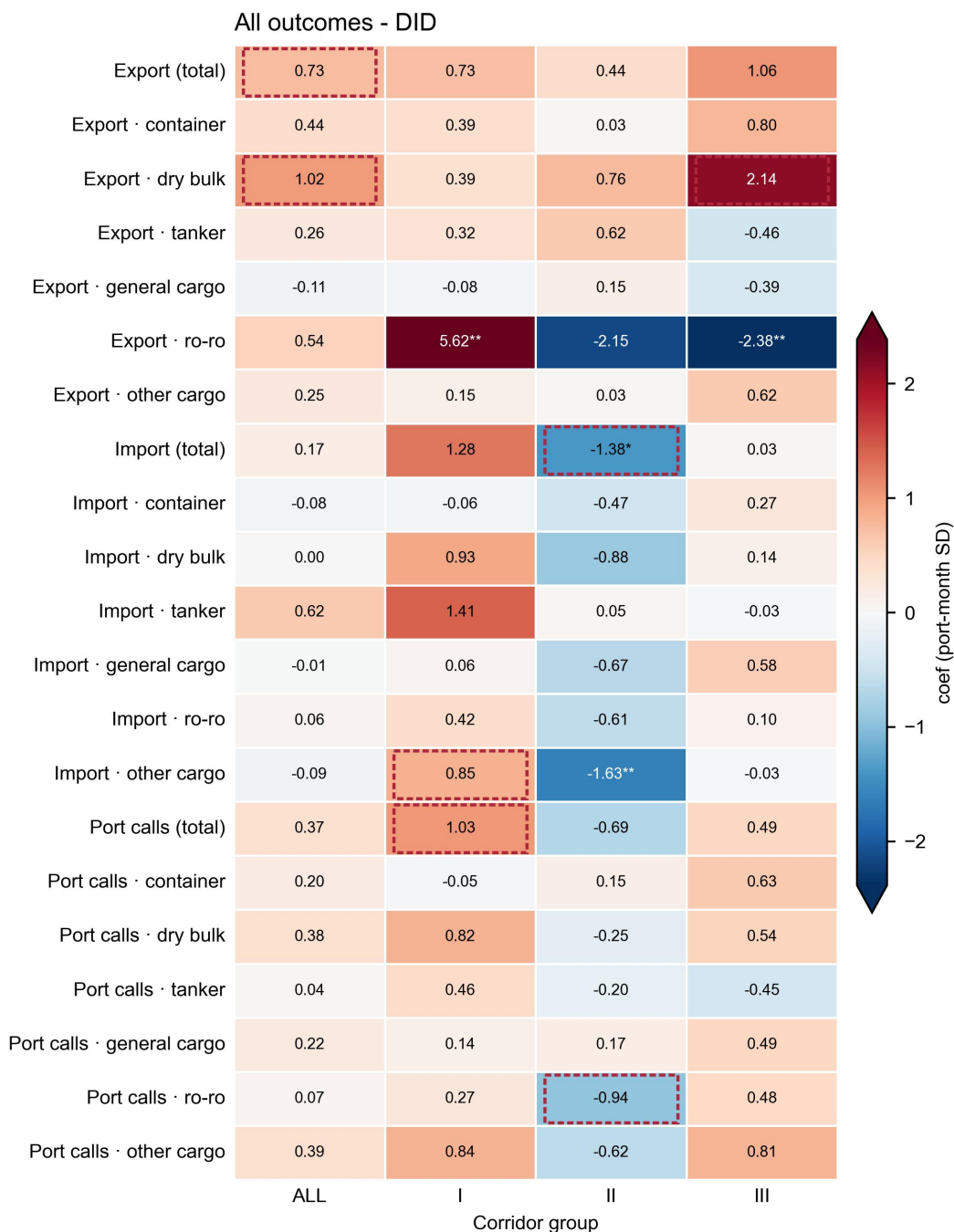


Fig. A.1. Full coefficient heatmap, difference-in-differences (DID). Estimated $E_{geo} \times post$ coefficients for all 21 outcomes — the three margins (exports, imports, port calls), each decomposed into total activity and six cargo/service types — across the four corridor groups. Cell values, asterisks, and dashed-red borders follow the conventions of Fig. 4, with the Benjamini–Hochberg correction applied within the full 84-test DID family; the color scale spans the 97th percentile of absolute coefficients.

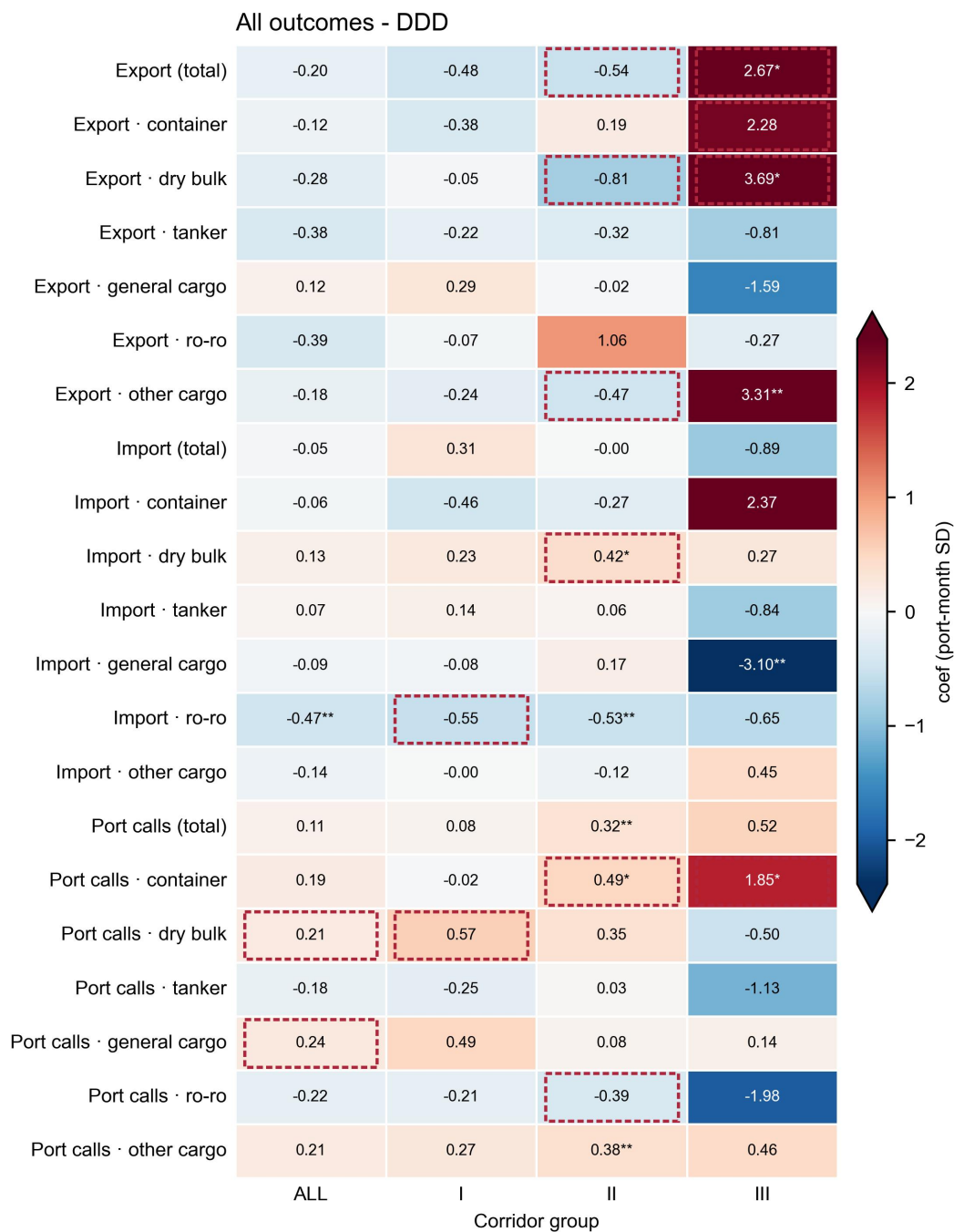


Fig. A.2. Full coefficient heatmap, connectivity-conditioned triple difference (DDD). Estimated $E_{geo} \times E_{net} \times$ post coefficients, with layout and conventions as in Fig. A.1.